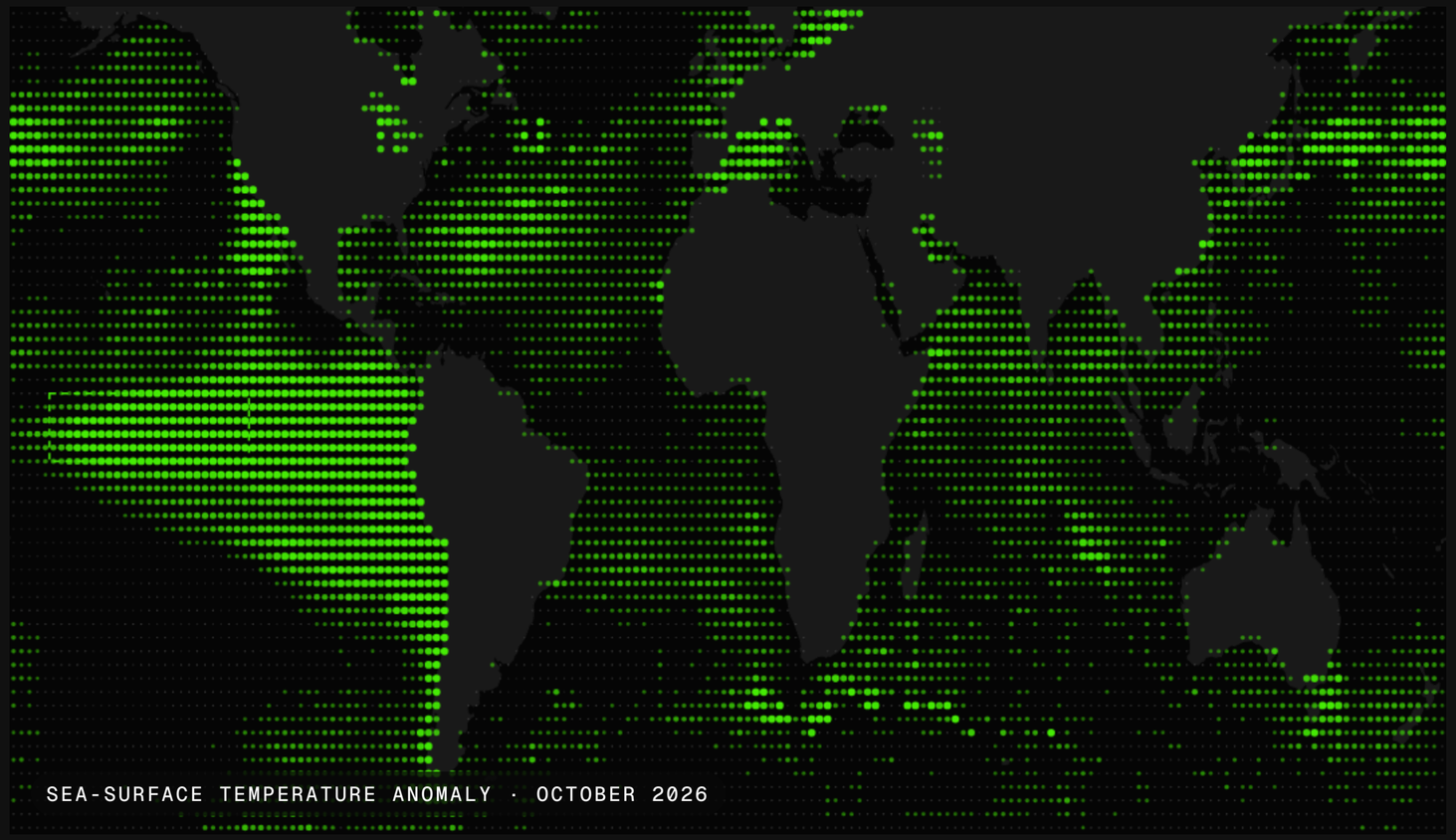
HRT — HORTUS · CLIMATE AND CROPS

hortus

The 2026–27 El Niño, read from open data.

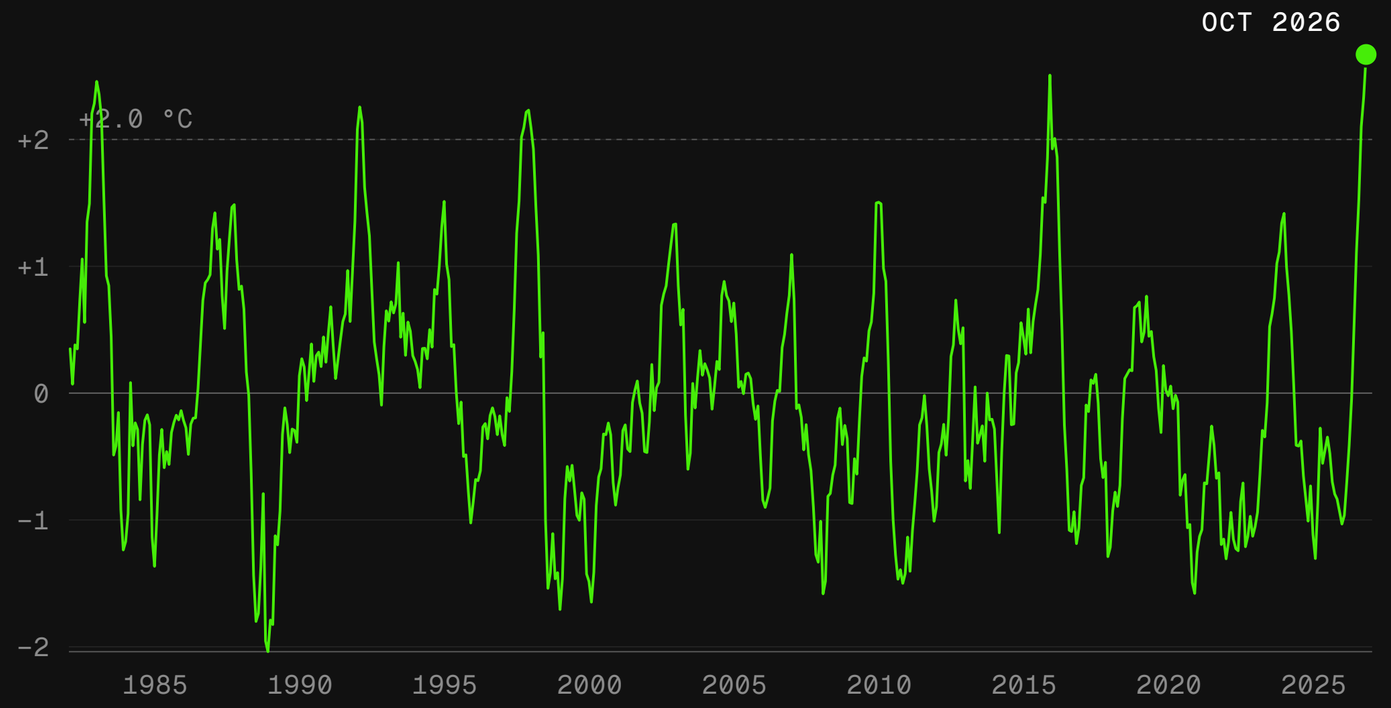
Where the harvest is at risk, measured from ocean temperatures, satellite rainfall and forty years of national yields.



The Pacific is already past every month since 1982.

+2.67 °C

Relative Niño-3.4, October 2026 (1–24 Oct, preliminary OISST). The previous highest month was +2.51 °C in November 2015.



DATA: NOAA OISST V2.1 · NASA POWER (IMERG, MERRA-2) · FAO VIA OUR WORLD IN DATA · WORLD BANK WDI

2026 is running ahead of 1982, 1997 and 2015.



2026 1982, 1997, 2015, 2023

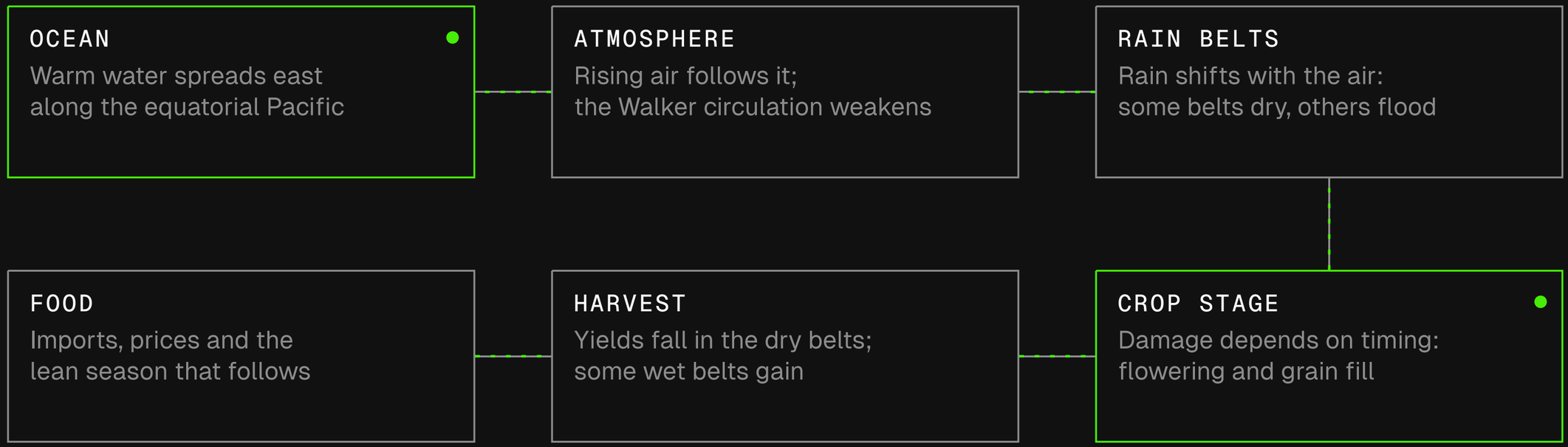
DATA: NOAA OISST V2.1 · NASA POWER (IMERG, MERRA-2) · FAO VIA OUR WORLD IN DATA · WORLD BANK WDI

Forecasters expect a strong to very strong event into early 2027.

NOAA	El Niño Advisory in force	STRONG TO VERY STRONG EVENT FAVOURED THROUGH JANUARY-MARCH 2027 (CPC, 8 OCT 2026)
RONI	A new official index	RELATIVE NIÑO-3.4, USED BY CPC FROM 1 FEB 2026, REMOVES THE TROPICAL-MEAN WARMING
FEWS NET	Highest concern	SOUTHERN AFRICA AND UNIMODAL EAST AFRICA, WITH NEEDS PEAKING IN THE 2027 LEAN SEASON

SOURCES: NOAA CPC ENSO DISCUSSION, 8 OCT 2026 · NWS NOTICE 26-05 · FEWS NET SPECIAL REPORT, OCT 2026

How El Niño reaches a field.



MECHANISM AFTER ROPELEWSKI & HALPERT (1987) AND IIZUMI ET AL. (2014)

Open data, nothing else.

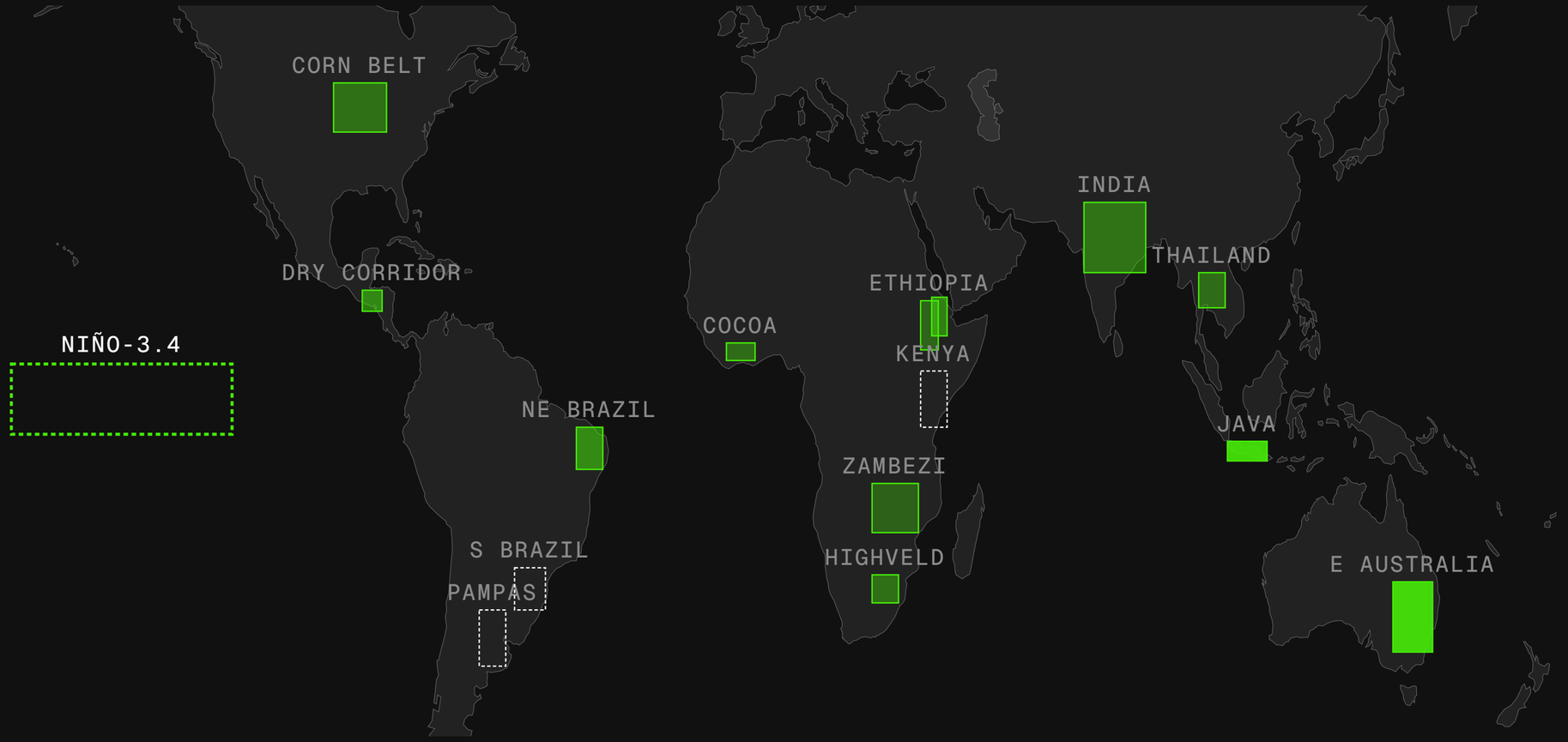
OCEAN	NOAA OISST v2.1	DAILY, 0.25°, 1982 TO 24 OCT 2026, THREE DAYS A MONTH
RAIN	NASA IMERG	SATELLITE AND GAUGE, 2001 TO SEP 2026
HEAT	NASA MERRA-2	DAILY MAXIMUM TEMPERATURE, 1982 TO SEP 2026
YIELD	FAO	CROP YIELDS 1961-2018; CEREAL YIELDS TO 2024

Method

Relative Niño-3.4 by the CPC method. Seasonal rainfall per crop belt. Yields detrended with robust LOWESS. Linear sensitivity per °C, 80% prediction intervals, robustness across six detrending methods.

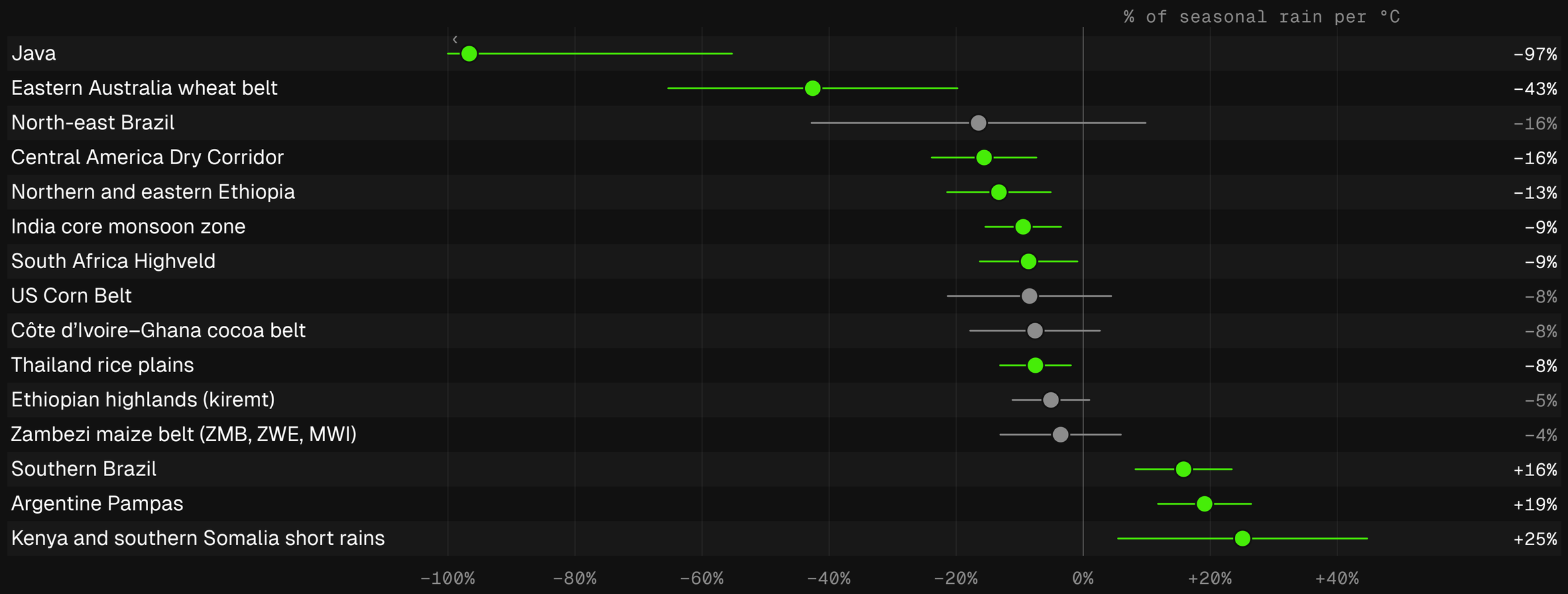
DATA: NOAA OISST V2.1 · NASA POWER (IMERG, MERRA-2) · FAO VIA OUR WORLD IN DATA · WORLD BANK WDI

Fifteen crop belts, from Java to the Pampas.



DATA: NOAA OISST V2.1 · NASA POWER (IMERG, MERRA-2) · FAO VIA OUR WORLD IN DATA · WORLD BANK WDI

Rain moves; it does not vanish.

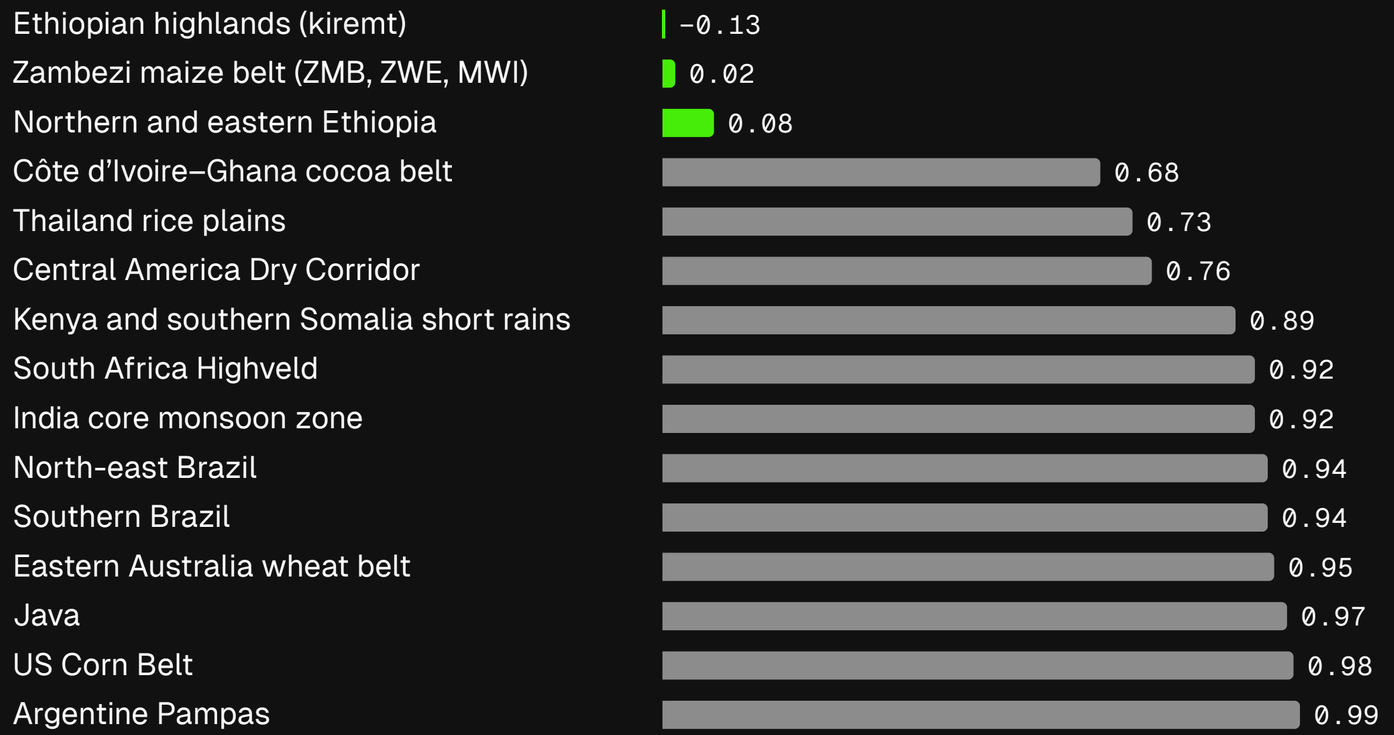


IMERG SEASONAL RAINFALL REGRESSED ON RELATIVE NIÑO-3.4, 2001–2025; BARS ARE 95% CONFIDENCE INTERVALS

One dataset missed Southern Africa’s 2024 drought.

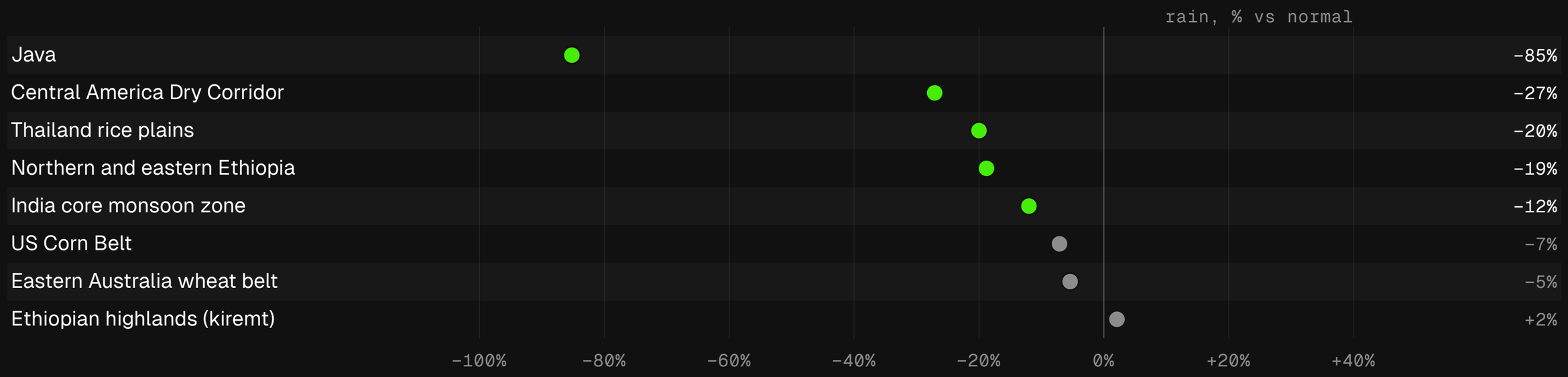
+52 pts

MERRA-2 put the 2023–24 Zambezi maize-belt rain near normal; IMERG measured –53%. Across Ethiopia and the Zambezi belt, the two agree almost not at all.



SEASON DEC 2023–MAR 2024 AND OTHERS: MERRA-2 MINUS IMERG, PERCENTAGE POINTS

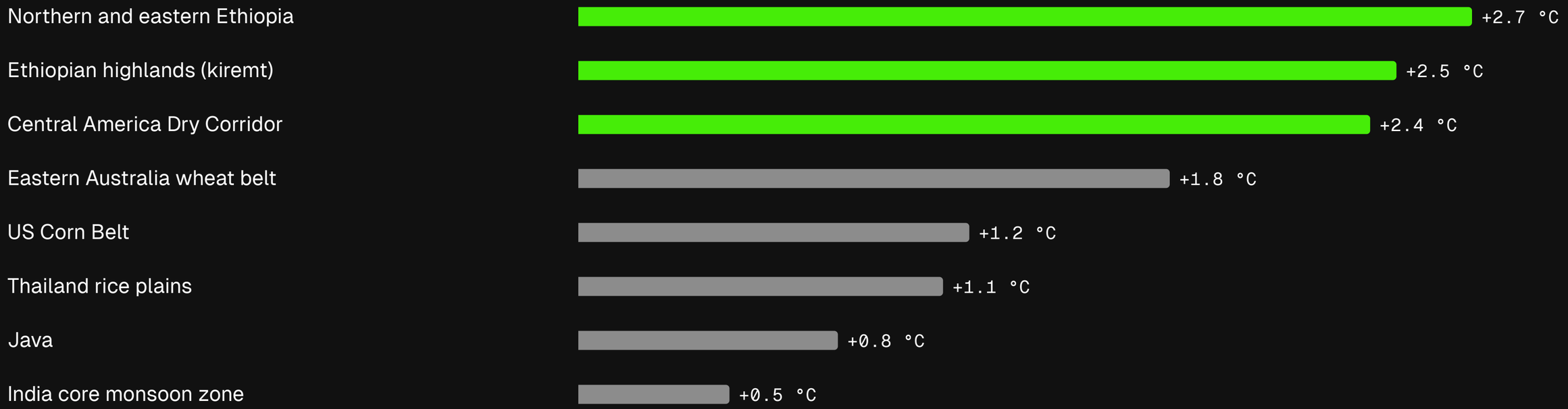
2026 so far: the dry side is already dry.



India’s weather service closed the June–September monsoon at 87% of its long-period average (–12.6%).

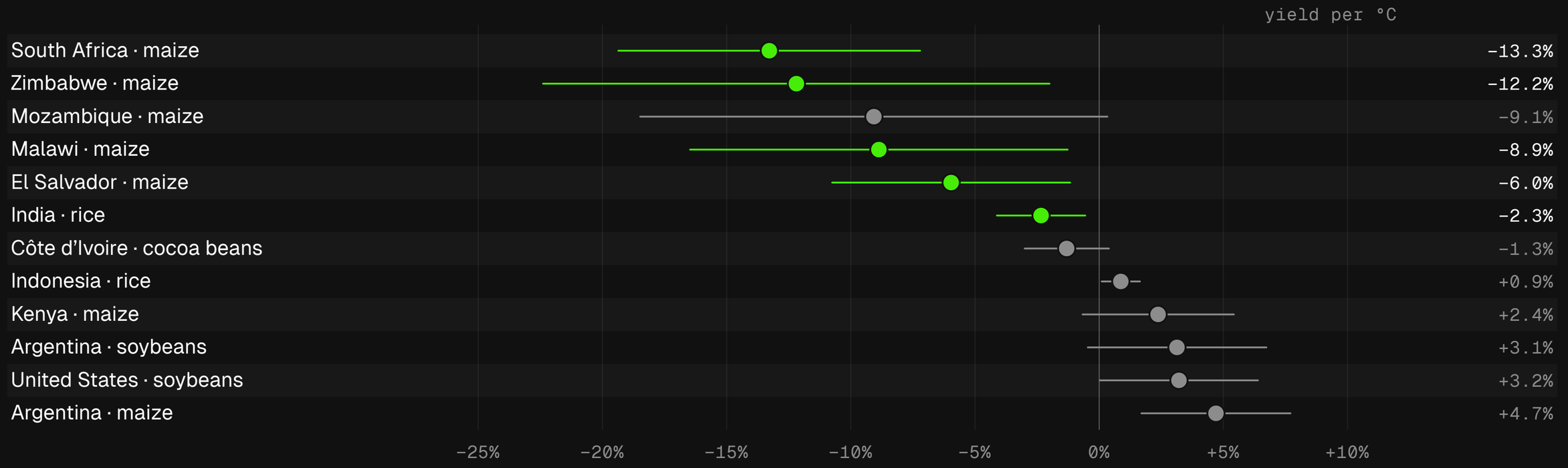
IMERG, SEASON TO DATE, % OF THE 2001-2020 MEAN FOR THE SAME MONTHS

And the heat came with it.



MERRA-2 MEAN DAILY MAXIMUM, SEASON TO DATE, VS 1991-2020

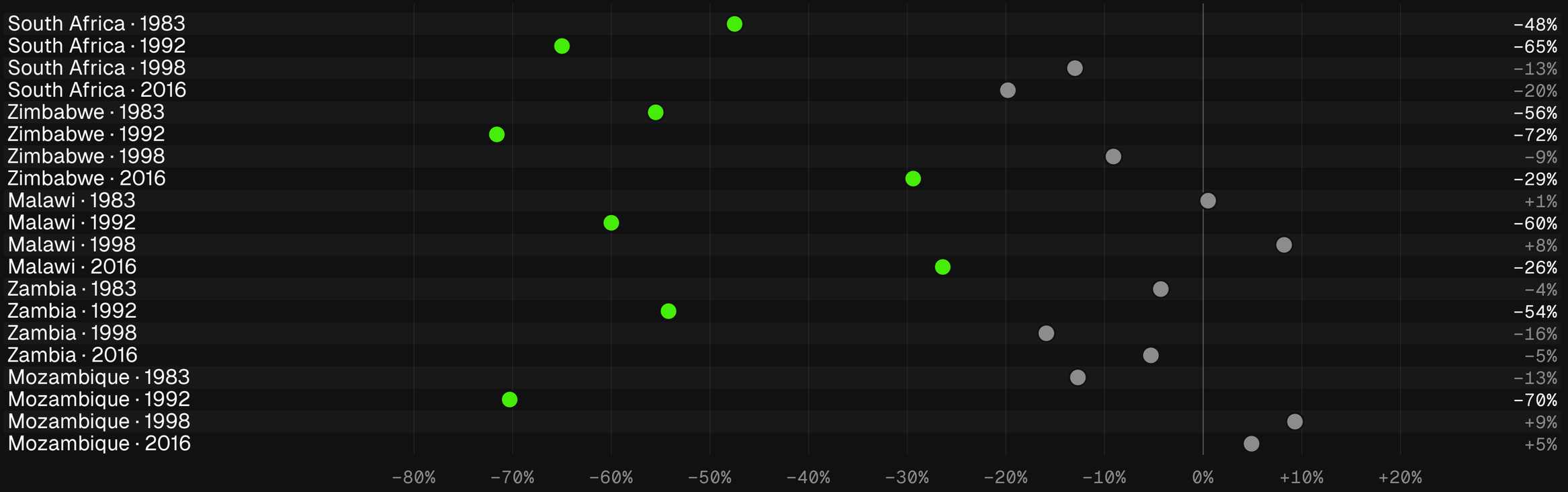
Southern Africa’s maize carries the largest signal.



The twelve crop–country pairs with p below 0.15; all 23 are in the paper.

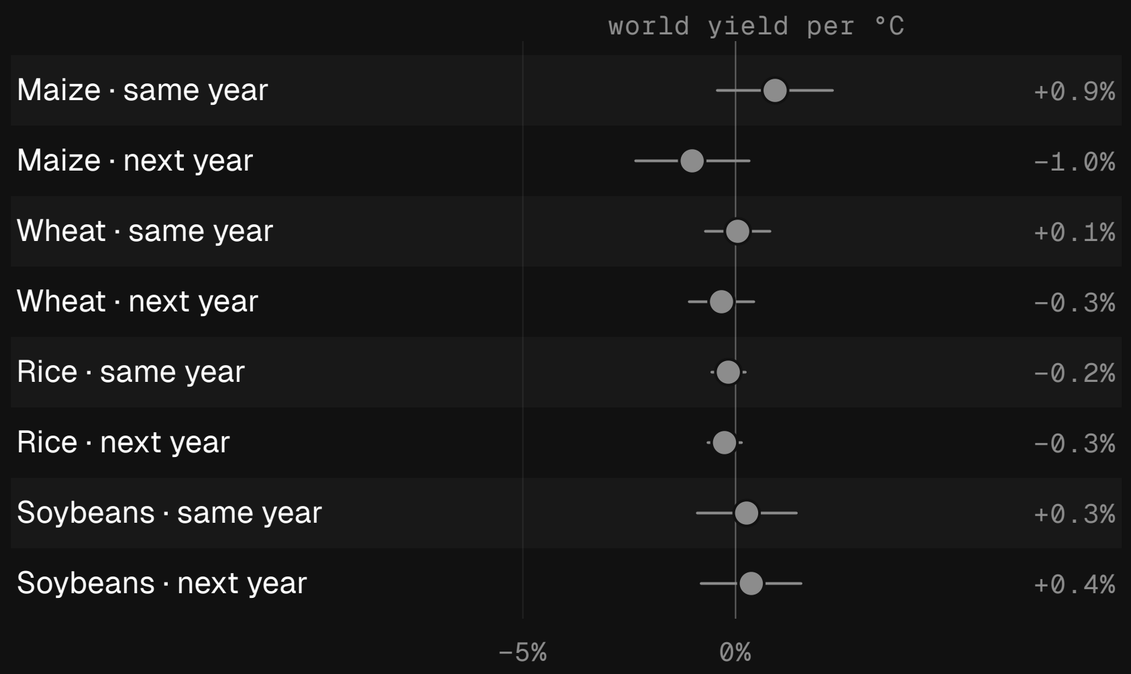
FAO YIELDS 1982-2018, DETRENDED; SLOPE PER °C OF THE CROP-SEASON RELATIVE NIÑO-3.4; 95% CI

After very strong events, 16 of 20 maize harvests fell short.



MAIZE YIELD VS TREND IN THE HARVEST AFTER EACH VERY STRONG EL NIÑO (1982, 1991, 1997, 2015)

World totals barely move.

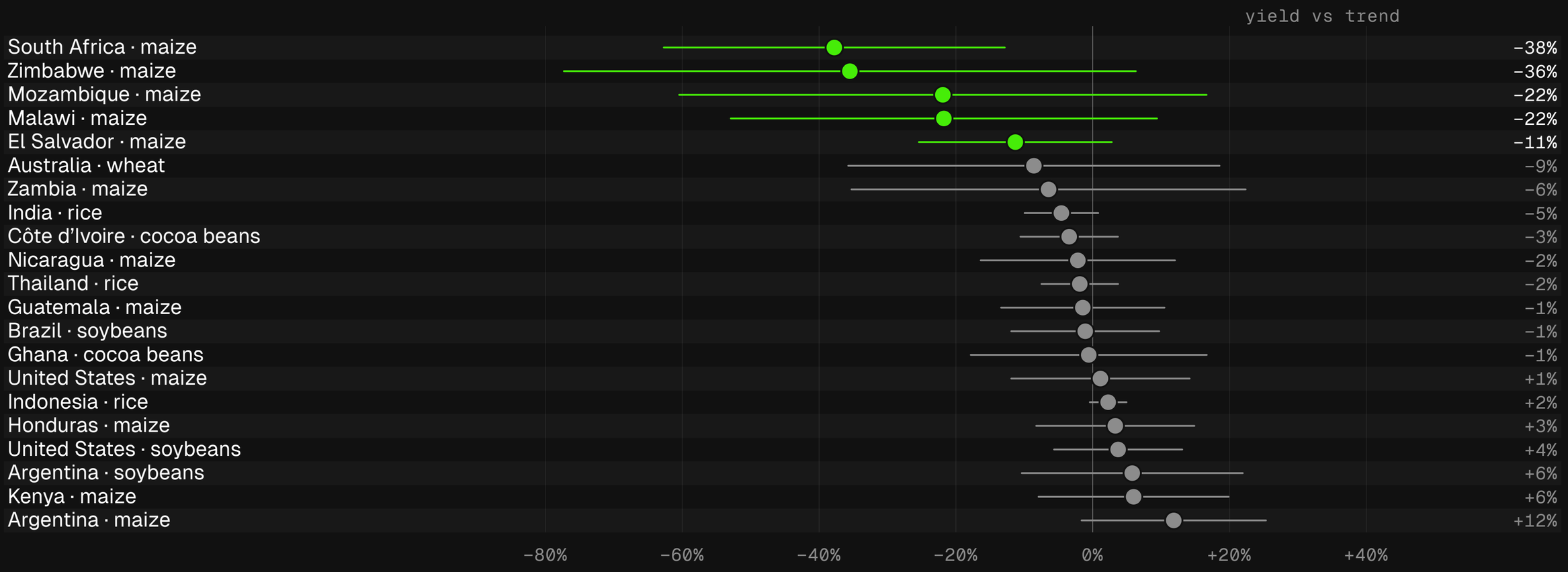


Global maize, wheat, rice and soybean yields show no significant response to El Niño strength (all p > 0.1). Losses in one belt are offset by gains or normal harvests elsewhere.

The global average hides the households that live inside the dry belts.

WORLD YIELDS, FAO; ANOMALIES IN THE HARVEST YEAR AFTER 1982, 1991, 1997 AND 2015

The outlook at 2.5 °C.



MEAN CHANGE VS TREND AT A CROP-SEASON INDEX OF 2.5 °C (OBSERVED INDEX FOR SEASONS UNDER WAY);
BARS: 80% PREDICTION INTERVALS

Where the risk concentrates.

CROP · COUNTRY	HARVEST	MEAN	80% RANGE	P(LOSS > 10%)
Maize · South Africa	2027	−38%	−63 to −13%	92%
Maize · Zimbabwe	2027	−36%	−77 to +6%	78%
Maize · Malawi	2027	−22%	−53 to +9%	69%
Maize · Mozambique	2027	−22%	−60 to +17%	66%
Maize · El Salvador	2026	−11%	−25 to +3%	55%
Wheat · Australia	2026	−9%	−36 to +19%	47%
Maize · Zambia	2027	−6%	−35 to +22%	44%
Maize · Nicaragua	2026	−2%	−16 to +12%	24%

PROBABILITY OF A YIELD LOSS LARGER THAN 10%, FROM THE SAME MODELS

Some farms gain.

+12%

ARGENTINA · MAIZE

Wetter season in El Niño years

+6%

ARGENTINA · SOYBEANS

Wetter season in El Niño years

+6%

KENYA · MAIZE

Wetter season in El Niño years

+4%

UNITED STATES · SOYBEANS

Season observed so far

The Pampas and southern Brazil usually get more rain in El Niño years; Kenya’s short rains too. Too much rain at harvest can still cost quality.

SAME MODELS; POSITIVE SENSITIVITY, HARVEST 2027 UNLESS NOTED

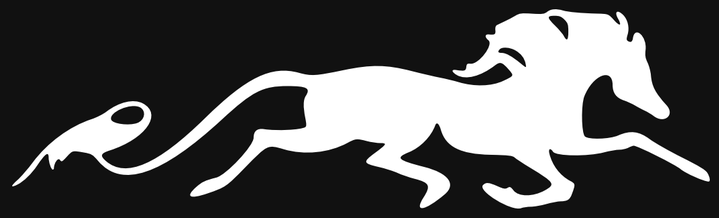
What to do now.

AGENCIES	Plan for Southern Africa’s 2027 lean season	PRE-POSITION MAIZE AND CASH SUPPORT; FEWS NET RATES CONCERN HIGH	STRONG
GROWERS	Shorter-season maize, water saved early	WHERE THE DRY BELTS FACE DECEMBER–MARCH PLANTING	MODERATE
BUYERS	White maize into the SADC region	REGIONAL TRADE CARRIES THE SHOCK BEYOND THE DRY FARMS	MODERATE
HORTUS	Update monthly	EVERY INPUT HERE REFRESHES WITHIN WEEKS OF THE MONTH’S END	METHOD

RECOMMENDATIONS FOLLOW FROM THE EVIDENCE ABOVE; STRENGTH OF EVIDENCE NOTED ON EACH

What this cannot tell you.

01	Beyond the record	A 2.5–3.0 °C season extrapolates past every year in the sample
02	Nations, not fields	National yields blur the belts that drive them
03	Yield is not supply	Area planted, stocks and trade decide what reaches a table
04	Other oceans	The Indian Ocean dipole can amplify or offset El Niño
05	Preliminary data	October SSTs and recent rain will be revised
06	Correlation	Linear slopes summarise history; they do not model crops



Nothing left to miss.

The full paper, the data steps and the live outlook are in the Hortus research pages.